Technical University of Dortmund

Department of Statistics

Statistical Analysis of Student Performance in Mathematics and Language

Application Report for the M.Sc. Data Science Program

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Submitted by: Treva Antony Ogwang

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1. Introduction

Student performance can be related to several background factors. In educational data analysis, it is useful to examine whether academic outcomes differ across groups such as gender or family educational background. Such analysis does not explain every reason behind student achievement, but it can identify patterns that may be relevant for further investigation.

This report analyzes the performance of 486 students in Mathematics and Language. The main aim is to examine whether scores differ by gender and by parental level of education. Two research questions guide the analysis. First, the report investigates whether male and female students differ significantly in their Mathematics and Language scores. Second, it examines whether scores differ across parental education levels and, if differences are found, which groups differ from each other.

To answer these questions, the analysis uses descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way analysis of variance, Tukey HSD post-hoc tests, and effect size measures. Descriptive statistics and figures are used to summarize the data, while hypothesis tests are used to assess whether the observed group differences are statistically significant.

The results show statistically significant gender differences in both subjects. Male students had a higher mean Mathematics score, while female students had a higher mean Language score. Significant differences were also found across parental education levels. Students whose parents had a high school level of education generally had lower mean scores than students whose parents had higher education levels. However, the effect sizes show that parental education explained only a limited proportion of the total variation in scores. Therefore, the findings should be interpreted as statistical associations, not as causal effects.

The report is structured as follows. Section 2 describes the dataset and research questions. Section 3 explains the statistical methods. Section 4 presents the descriptive results. Sections 5 and 6 report the gender and parental education results, and Section 7 summarizes the findings and limitations.

2. Data and Research Questions

The dataset used in this report contains information on student performance in two subjects, Mathematics and Language. According to the task description, both scores are measured on a scale from 0 to 100. The dataset also includes gender and parental level of education. The variable gender is nominal, while parental level of education is an ordinal categorical variable with four levels: high school, associate's degree, bachelor's degree, and master's degree. Mathematics and Language scores are numerical variables.

The original dataset was stored in long format. This means that each student appeared in two separate rows, one row for the Mathematics score and one row for the Language score. The dataset initially contained 972 rows. A check of the student identifiers showed that each of the 486 students appeared exactly twice and had exactly two subject records. Therefore, the data were reshaped into wide format, so that each student formed one observation with separate columns for Mathematics and Language scores.

After reshaping, the final analysis dataset contained 486 observations and five variables: student identifier, gender, parental level of education, Mathematics score, and Language score. No missing values were found in the variables used for the analysis. All observed scores were within the valid range from 0 to 100. The final dataset contained 245 female students and 241 male students.

The analysis is guided by two research questions. The first research question asks whether male and female students differ significantly in their Mathematics and Language scores. The second research question asks whether scores differ across parental education levels and, if differences are present, which parental education groups differ from each other.

For the first research question, Mathematics and Language scores are compared between female and male students. For the second research question, Mathematics and Language scores are compared across the four parental education groups. This structure allows the analysis to answer both whether group differences exist and where those differences occur.

3. Statistical Methods

The statistical methods were selected according to the research questions and the scale level of the variables. Mathematics and Language scores are numerical outcome variables. Gender has two independent groups, while parental level of education has four ordered categories. Therefore, the analysis used descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way ANOVA, Tukey HSD post-hoc tests, and effect size measures. All tests used a significance level of $\alpha = 0.05$.

3.1. Descriptive Statistics and Statistical Graphics

Descriptive statistics were used to summarize the Mathematics and Language scores before hypothesis testing. For each score variable, the number of observations, mean, median, standard deviation, minimum, and maximum were calculated. The mean was used to describe the location of the scores, while the standard deviation was used to describe their dispersion (Illowsky & Dean, 2013).

3.2. Welch Two-Sample t-Test

Welch's two-sample t-test was used to test whether female and male students differed significantly in Mathematics and Language scores. It is appropriate because gender has two independent groups and the score variables are numerical. Welch's version was chosen because it does not require equal variances between the two groups (Welch, 1947). The Welch t-statistic is defined as:

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\overline{x_1}$ and $\overline{x_2}$ are the group means, s_1^2 and s_2^2 are the group variances, and $n_1$and n_2 are the group sizes. In this report, group 1 refers to female students and group 2 refers to male students. The null hypothesis states that the female and male mean scores are equal.

3.3. Cohen's d

Cohen's d was used to measure the size of the gender differences. This is useful because a p-value shows whether a difference is statistically significant, but not how large the difference is. Cohen's d expresses the mean difference in standard deviation units (Cohen, 1988):

$$d = \frac{\overline{x_1} - \overline{x_2}}{s_p} \qquad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

In this report, $n_1 = 245$ is the number of female students and $n_2 = 241$ is the number of male students. A negative value of d means that the female mean is lower than the male mean, while a positive value means that the female mean is higher.

3.4. One-Way ANOVA, Tukey HSD, and Eta-Squared

One-way ANOVA was used to test whether Mathematics and Language scores differed across the four parental education levels. This method is appropriate because parental education has more than two groups and the outcome variables are numerical. ANOVA compares variation between groups with variation within groups (Illowsky & Dean, 2013):

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

Here, MS_{between} measures variation between the mean scores of the parental education groups, while MS_{within} measures variation among students within the same group. The null hypothesis states that all parental education group means are equal. A significant ANOVA result means that at least one group mean differs, but it does not show which specific groups differ.

If the ANOVA p-value was below the significance level of $\alpha = 0.05$, the result was treated as statistically significant. In that case, Tukey's HSD post-hoc test was used to compare pairs of parental education groups while controlling the family-wise error rate (Tukey, 1949). The basic Tukey HSD statistic is written as

$$HSD = q_{\alpha,k,N-k} \cdot \sqrt{\frac{MS_{\text{within}}}{n}}$$

where $q_{\alpha,k,N-k}$ is the critical value from the studentized range distribution, k is the number of groups, N is the total sample size, and n is the group size. Since the parental education groups had unequal sample sizes, the software implementation used an adjusted version of Tukey's procedure.

Eta-squared was used as an effect size for the ANOVA results. It measures the proportion of total score variation associated with parental education level (Cohen, 1988):

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

where SS_{between} is the variation explained by parental education groups and SS_{total} is the total variation in scores.

3.5. Assumption Checks

The t-tests and ANOVA assume independent observations, numerical outcome variables, approximate normality, and acceptable variance homogeneity. Independence was addressed by reshaping the dataset so that each student appeared once in the final analysis dataset. Histograms were inspected to assess the score distributions, and Levene's test was used to assess equality of variances across gender and parental education groups.

All analyses were conducted in Python (Python Software Foundation, 2026). The dataset was prepared and reshaped using pandas (McKinney, 2010). Welch t-tests and Levene's tests were conducted using SciPy (Virtanen et al., 2020), while one-way ANOVA and Tukey HSD post-hoc tests were conducted using statsmodels (Seabold & Perktold, 2010). The statistical graphics were created using matplotlib and seaborn (Hunter, 2007; Waskom, 2021).

4. Descriptive Results

Both subjects had complete data for all 486 students. The mean score was 66.38 in Mathematics and 69.00 in Language, while the median was 66.00 and 70.00, respectively. Mathematics scores were slightly more spread out, with a standard deviation of 15.05 compared with 14.45 for Language. All values were within the valid range from 0 to 100. The observed scores ranged from 23.00 to 100.00 in Mathematics and from 19.50 to 100.00 in Language.

Figure 1 shows the mean scores by gender and subject. Male students had a higher mean Mathematics score than female students, 68.95 compared with 63.86. For Language, the pattern was reversed, with female students scoring 73.00 compared with 64.93 for male students.

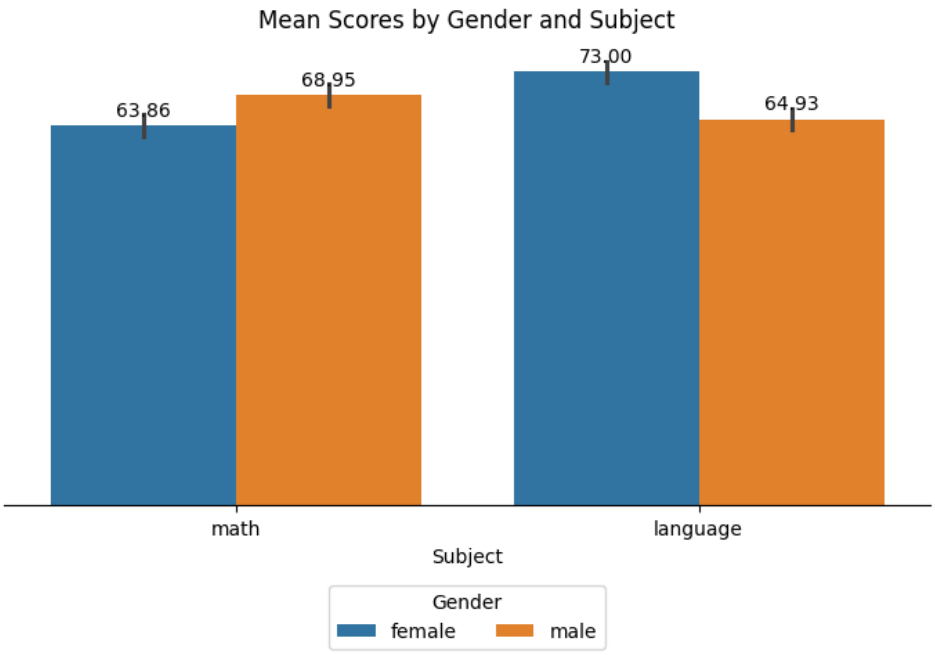


Figure 1: Mean scores by gender and subject

5. Gender Differences in Mathematics and Language

Welch two-sample t-tests were used to examine whether female and male students differed significantly in Mathematics and Language scores. For Mathematics, the Welch t-test showed a statistically significant difference, $t(483.23) = -3.78$, $p < .001$. Female students had a lower mean Mathematics score of 63.86, while male students had a mean score of 68.95. The effect size was $d = -0.343$, indicating a small to moderate difference.

For Language, the Welch t-test also showed a statistically significant difference, $t(484.00) = 6.41$, $p < .001$. Female students had a mean Language score of 73.00, while male students had a mean score of 64.93. The effect size was $d = 0.581$, indicating a moderate difference. These results answer the first research question: male students scored significantly higher in Mathematics, while female students scored significantly higher in Language.

6. Parental Education Differences in Mathematics and Language

One-way ANOVA was used to examine whether Mathematics and Language scores differed across parental education levels. Figure 2 shows that, in both subjects, the mean scores were lowest in the high school parental education group and highest in the master's degree group.

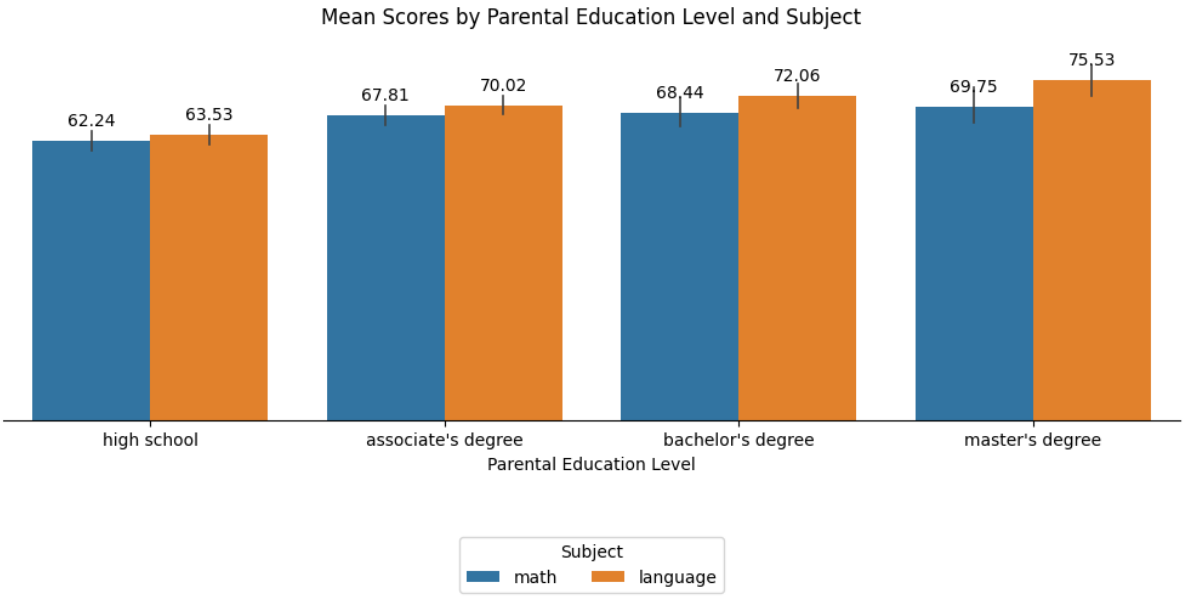


Figure 2: Mean scores by parental education level and subject with 95% confidence intervals

The ANOVA results were statistically significant for Mathematics, $F(3,482) = 6.22$, $p < .001$, $\eta^2 = 0.037$, and Language, $F(3,482) = 14.24$, $p < .001$, $\eta^2 = 0.081$. Thus, parental education explained about 3.7% of the variation in Mathematics scores and 8.1% of the variation in Language scores. Since both ANOVA results were significant, Tukey HSD tests were used to identify the differing groups. Table 1 shows the significant comparisons.

Subject	Group 1	Group 2	Mean difference	Adjusted p-value
Mathematics	Associate's degree	High school	-5.5700	0.0037
Mathematics	Bachelor's degree	High school	-6.1926	0.0079
Mathematics	High school	Master's degree	7.5022	0.0055
Language	Associate's degree	High school	-6.4905	0.0001
Language	Associate's degree	Master's degree	5.5028	0.0429
Language	Bachelor's degree	High school	-8.5265	< .001
Language	High school	Master's degree	11.9934	< .001

Table 1: Significant Tukey HSD post-hoc comparisons

The Tukey HSD results show that the high school parental education group differed significantly from the higher parental education groups in both subjects. In Mathematics, no significant differences were found among the associate's, bachelor's, and master's degree groups. In Language, the master's degree group also scored significantly higher than the associate's degree group. Overall, the parental education differences were mainly driven by lower scores in the high school group, while eta-squared values showed that parental education explained only a limited proportion of total score variation.

7. Summary and Discussion

This report analyzed whether student performance in Mathematics and Language differed by gender and parental level of education. The analysis used descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way ANOVA, Tukey HSD post-hoc tests, and effect size measures on a dataset of 486 students.

The results showed statistically significant gender differences in both subjects. Male students had a higher mean Mathematics score, while female students had a higher mean Language score. The effect size was small to moderate for Mathematics and moderate for Language, suggesting that the gender difference was more pronounced in Language.

The results also showed statistically significant differences across parental education levels for both subjects. Tukey HSD tests showed that the high school parental education group generally had lower scores than the higher parental education groups. For Language, the master's degree group also differed significantly from the associate's degree group.

These findings should be interpreted carefully. Parental education explained about 3.7% of the variation in Mathematics scores and about 8.1% in Language scores, so it was associated with performance but did not explain most score variation. The results are statistical associations, not causal effects. Other factors such as study habits, school quality, prior achievement, socioeconomic conditions, teacher support, and language background may also influence student performance.

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